Radio & Digital Ecosystem Setup

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Public Technical Overview

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Public Version

System Overview - Communications Architecture

Mission

- Integrated VoIP, RF, SDR, and satellite platform
- Designed for resilience and interoperability

Network Architecture

- Segmented network design
- Secure VPN mesh overlay
- Primary WAN + failover WAN

Security & Controls

- SIP auth + firewall
- Channel limit + spend control
- Repeat call guard
- Guest network isolation

Core Capabilities

- Asterisk + SIP Provider
- AllStar Node XXXXX
- DVSwitch + OpenSpot4 Pro
- ADS-B, AIS, spectrum SDR

Cloud & Resiliency

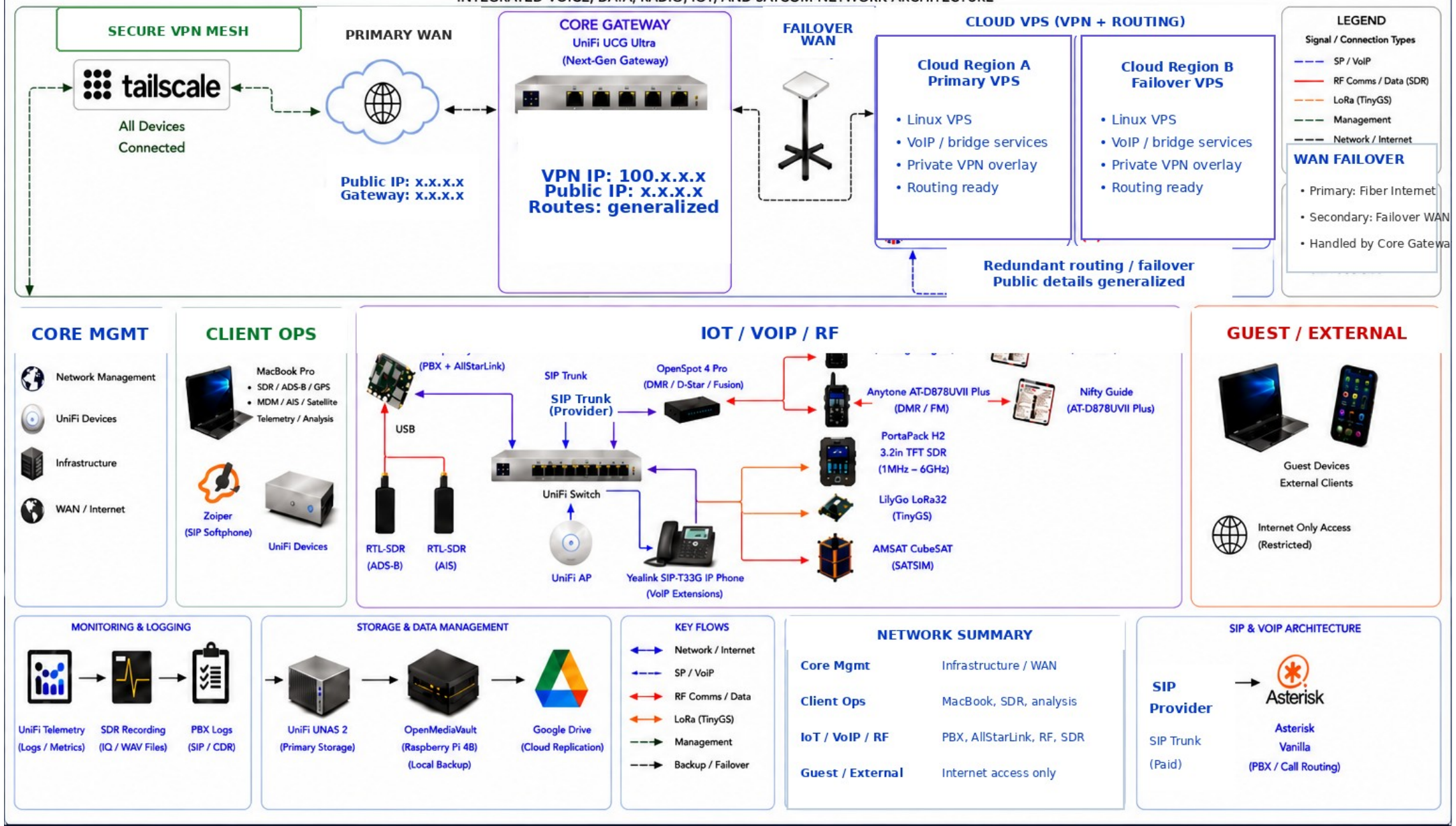
- Cloud Region A (Asterisk)
- Cloud Region B (DVSwitch)
- IPv6 / BGP failover

Validation

- VoIP, RF, and network tested
- Device-level validation
- Test plan + report tracking

VOIP + RADIO + IOT NETWORK – TACTICAL COMMUNICATIONS ECOSYSTEM

INTEGRATED VOICE, DATA, RADIO, IOT, AND SATCOM NETWORK ARCHITECTURE



All existing equipment (owned) shown in GREEN – $0 cost. Items in RED are to acquire (one-time cost) and/or have monthly service fees.

Test Objectives – VoIP

- Validate Zoiper internal extensions
- Confirm SIP Provider inbound/outbound routing
- Verify SIP audio, DTMF, routing

Test Objectives – RF & Bridge

- Validate AllStar Node XXXXX
- Confirm Zoiper/IAX access
- Verify DVSwitch connectivity
- Test DMR / Fusion routing

Test Objectives – Devices & Network

- Validate OpenSpot + radios
- Confirm SDR (ADS-B / AIS)
- Verify Tailscale connectivity
- Validate VLAN routing + failover

Test Objectives – Security

- Confirm SIP auth + firewall
- Verify spend limits + call guard
- Ensure RTP security
- Validate logging + uptime

Test Plan – VoIP

- Dial 1000/1001 → 2-way audio
- Call XXXX DID → inbound works
- Dial 9+number → outbound works

Test Plan – AllStar / DVSwitch

- Dial Rpt(XXXXX,P)
- Verify RF TX/RX audio
- Confirm DVSwitch bridge

Test Plan – Radios / SDR

- Test Yaesu - FT-70DR / Anytone - AT-D878UVII Plus
- Verify OpenSpot connection
- Validate ADS-B / AIS / PortaPack/AMSAT CubeSATSIM
- End-to-end validation:
- Radio → OpenSpot → DVSwitch → AllStar → Asterisk → SIP Provider → PSTN

Test Plan – Network / Security

- Test Tailscale connectivity
- Validate VLAN routes
- Test failover Fiber Internet → Failover WAN
- Verify call guard + spend limit

Device-Level Validation Coverage

Core Network

- Gateway routing + WAN
- Switch + AP operational
- Primary + failover WAN
- Network segmentation

VoIP / SIP

- SIP provider trunks
- SIP registration + routing
- Asterisk CDR logging

Digital Bridge

- DVSwitch service running
- AllStar ↔ DVSwitch ↔ OpenSpot
- DMR / Fusion routing

Security

- Guest network isolated
- SIP auth + firewall
- Call guard + spend limits

Cloud / VPS

- Cloud Region A reachable
- Cloud Region B reachable
- IPv6 / BGP validation

RF / Radio

- AllStar Node XXXXX
- OpenSpot4 Pro connectivity
- FT-70DR / AT-D878UVII Plus

SDR / IoT

- ADS-B + AIS decoding active
- PortaPack spectrum analysis
- TinyGS (LoRa telemetry)
- CubeSatSim validation

Test Report – Summary

- Overall Status: PASS / FAIL
- Subsystem validation complete
- Issues documented

Test Report – Key Checks

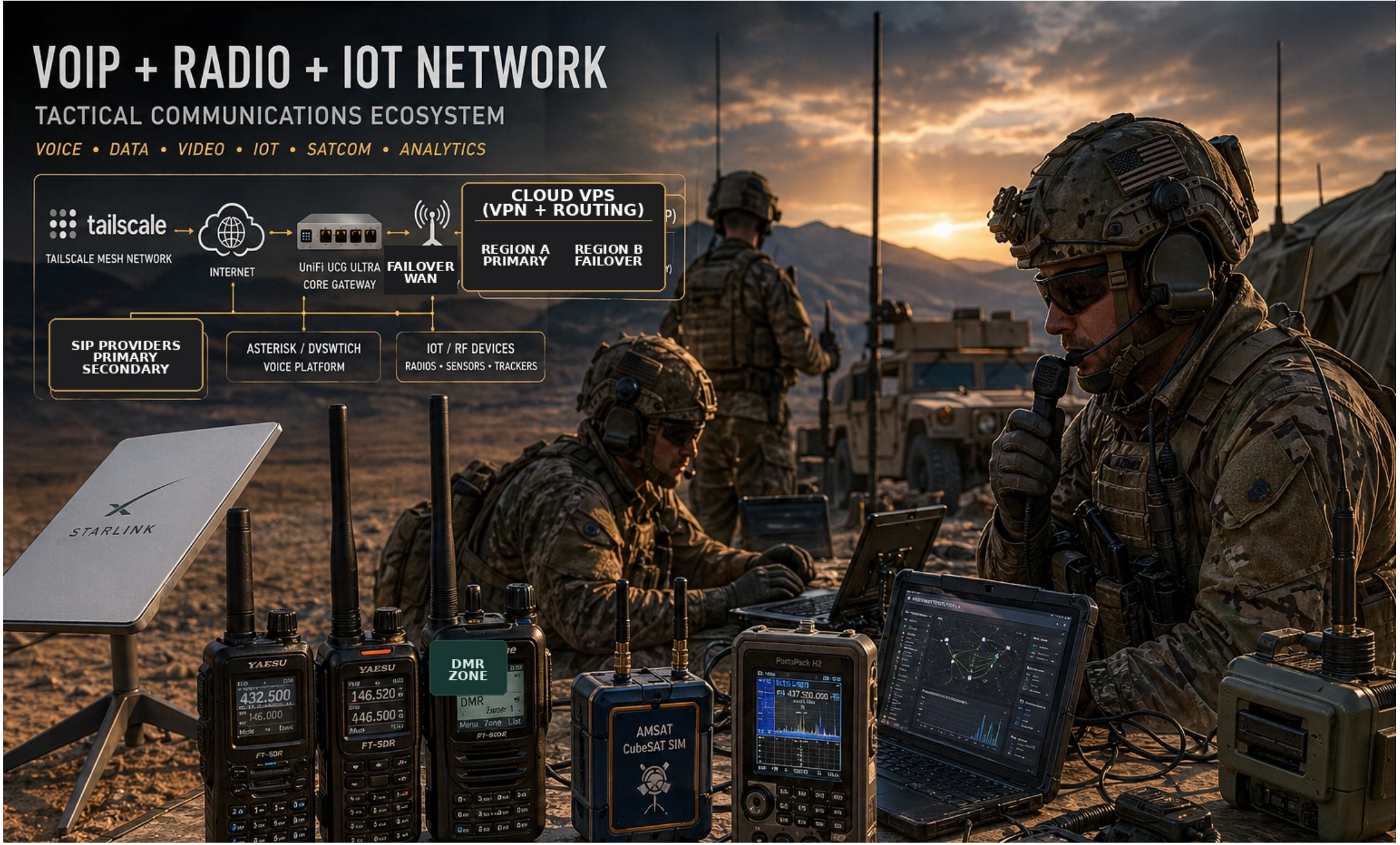
- VoIP: PASS / FAIL
- AllStar: PASS / FAIL
- DVSwitch: PASS / FAIL
- Radios: PASS / FAIL
- Network: PASS / FAIL
- SDR/IoT: Pass/FAIL

VOIP + RADIO + IOT NETWORK - PUBLIC BILL OF MATERIALS

Public overview; vendor and infrastructure identifiers generalized.

LEGEND
$0 = Existing
$ = To Acquire

CATEGORY	ITEM #	ITEM / MODEL	DESCRIPTION / ROLE	TYPE	QTY	STATUS	EST. UNIT COST	EST. TOTAL COST
NETWORK	1	SIP Provider	VoIP trunk provider for inbound / outbound calling	Service	1	To Acquire	$3.00 / mo	$3.00 / mo
	2	VPN Mesh	Secure mesh VPN for remote access	Service	1	Existing	$0	$0
	3	Core Gateway	Gateway / firewall / network controller	Hardware	1	Existing	$0	$0
	4	Primary Internet	Primary wired internet connection	Service	1	Existing	$0	$0
	5	Failover WAN Kit	Secondary satellite / wireless failover path	Hardware	1	To Acquire	$500.00	$500.00
	6	Cloud VPS (Primary)	Primary cloud services host	Service	1	Existing	$0	$0
	7	Cloud VPS (Secondary)	Secondary cloud redundancy / failover host	Service	1	Existing	$0	$0
RF / RADIO	8	FT-70DR	Dual-band analog / digital handheld radio	Radio	1	Existing	$0	$0
	9	AT-D878UVII Plus	DMR / FM dual-band handheld radio	Radio	1	Existing	$0	$0
	10	OpenSpot 4 Pro	DMR / D-Star / Fusion hotspot	Gateway	1	Existing	$0	$0
	11	DVSwitch	Digital voice switching / bridging	Software	1	Existing	$0	$0
	12	AllStar Node XXXXX	AllStarLink node (VoIP / RF bridge)	Node	1	Existing	$0	$0
IOT / SDR	13	PortaPack H2 SDR	RF analysis, capture, and spectrum monitoring	SDR	1	To Acquire	$300.00	$300.00
	14	TinyGS / LoRa32	LoRa telemetry device for IoT / remote sensors	IoT	1	Existing	$0	$0
	15	CubeSatSim Kit	CubeSat telemetry and communications simulator	Simulator	1	To Acquire	$550.00	$550.00
VOIP	16	IP Phone	Gigabit IP phone for VoIP extensions	Phone	1	To Acquire	$89.00	$89.00
	17	Softphone Client	SIP softphone client for laptop	Software	1	To Acquire	$49.95	$49.95
CLIENT / OPS	18	Laptop Workstation	Client for SDR, analysis, and operations	Computer	1	Existing	$0	$0
	19	Raspberry Pi 4B	PBX + AllStarLink controller	Compute	1	Existing	$0	$0
	20	Network Switch	Switch for IoT / VoIP / RF devices	Network	1	Existing	$0	$0
	21	Access Point	Wi-Fi access point	Network	1	Existing	$0	$0


VOIP + RADIO + IOT NETWORK
TACTICAL COMMUNICATIONS ECOSYSTEM
VOICE • DATA • VIDEO • IOT • SATCOM • ANALYTICS
tailscale
TAILSCALE MESH NETWORK
INTERNET
UniFi UCG ULTRA
CORE GATEWAY
FAILOVER WAN
CLOUD VPS (VPN + ROUTING)
REGION A PRIMARY
REGION B FAILOVER
SIP PROVIDERS PRIMARY SECONDARY
ASTERISK / DVSWTICH
VOICE PLATFORM
IOT / RF DEVICES
RADIOS • SENSORS • TRACKERS
STARLINK
YAESU
432.500
146.520
446.500
FT-5DR
DMR ZONE
AMSAT CubeSAT SIM
PortaPack H2